

RINCHINHUMBE, Mongolia

WMO: 442030

Lat: 51.1141N

Lon: 99.6681E

Elev: 5194

StdP: 12.14

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-43.9	-40.4	-48.5	0.4	-43.7	-45.1	0.5	-40.2	20.8	-3.9	14.7	-2.8	0.5	110	0.744

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	77.4	56.8	73.8	55.4	70.7	54.3	59.5	71.5	57.8	69.3	56.3	67.2	4.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
55.3	79.1	63.2	53.3	73.5	61.7	51.6	68.8	60.4	28.6	71.0	27.3	69.6	26.2	67.0	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4)	21.3	17.7	14.8	DB	-47.5	84.0	4.3	3.5	-50.6	86.5	-53.1	88.6	-55.6	90.5	-58.7	93.0
(5)				WB	-47.9	63.6	4.1	2.9	-50.8	65.6	-53.2	67.3	-55.5	68.9	-58.5	71.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	20.9	-26.0	-15.7	5.4	29.6	42.2	53.9	58.4	54.0	41.9	24.7	-0.5	-19.8	(6)
(7)		DBStd	30.98	9.27	10.68	12.48	9.59	8.05	6.55	4.97	5.72	7.46	10.19	13.05	10.55	(7)
(8)		HDD50	11239	2356	1839	1382	612	264	37	3	27	256	785	1514	2164	(8)
(9)		HDD65	16122	2821	2259	1847	1062	708	335	214	341	692	1250	1964	2629	(9)
(10)		CDD50	602	0	0	0	0	21	153	262	152	14	0	0	0	(10)
(11)		CDD65	10	0	0	0	0	0	1	8	1	0	0	0	0	(11)
(12)		CDH74	295	0	0	0	0	8	81	151	54	1	0	0	0	(12)
(13)		CDH80	41	0	0	0	0	0	5	30	6	0	0	0	0	(13)
(14)	Wind	WSAvg	3.1	1.0	1.3	2.7	5.3	5.9	4.3	3.5	3.1	3.5	2.9	1.8	1.2	(14)
(15)	Precipitation	PrecAvg	10.7	0.1	0.1	0.1	0.2	0.7	2.0	3.2	2.6	1.1	0.3	0.2	0.2	(15)
(16)		PrecMax	15.5	0.2	0.2	0.6	1.0	1.7	4.5	5.9	4.4	2.9	0.8	0.6	0.5	(16)
(17)		PrecMin	7.2	0.0	0.0	0.0	0.0	0.0	0.4	1.3	1.3	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.1	0.1	0.1	0.1	0.2	0.4	0.9	1.2	0.9	0.7	0.2	0.1	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	16.2	42.5	61.4	74.4	80.3	83.8	79.3	70.9	54.1	33.8	13.5	(19)
(20)		0.4%	MCWB	5.7	13.9	33.9	43.7	51.3	55.6	60.1	58.0	52.2	41.0	28.8	11.5	(20)
(21)		2%	DB	-1.1	9.3	35.5	54.5	68.9	76.4	78.3	74.3	65.3	48.4	28.4	6.1	(21)
(22)		2%	MCWB	-2.2	7.5	29.1	39.5	48.7	54.9	58.7	56.5	49.4	37.9	24.7	5.0	(22)
(23)		5%	DB	-7.7	4.4	29.9	49.5	63.9	72.5	74.2	70.6	60.8	43.8	23.9	0.3	(23)
(24)		5%	MCWB	-8.3	3.1	25.4	37.0	45.9	53.2	57.2	55.4	47.5	35.2	20.8	-0.6	(24)
(25)		10%	DB	-12.9	-0.1	24.9	45.1	59.5	69.0	70.7	67.2	56.2	39.7	18.8	-5.7	(25)
(26)		10%	MCWB	-13.4	-1.0	21.1	34.8	43.9	52.4	55.7	54.2	44.8	32.6	16.6	-6.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	14.0	34.0	44.2	52.3	59.3	63.0	60.5	53.8	42.0	28.9	11.8	(27)
(28)		0.4%	MCDWB	7.7	16.4	41.8	59.3	71.9	73.0	77.2	72.5	66.5	53.5	33.0	13.6	(28)
(29)		2%	WB	-2.1	7.7	29.3	40.1	49.5	56.9	60.6	58.4	50.6	38.8	24.8	5.0	(29)
(30)		2%	MCDWB	-1.2	9.3	35.1	52.7	66.3	70.4	73.6	70.4	62.7	48.1	28.2	6.1	(30)
(31)		5%	WB	-8.3	3.1	25.6	37.6	46.9	55.3	58.8	56.7	48.3	36.0	20.9	-0.5	(31)
(32)		5%	MCDWB	-7.7	4.2	29.9	48.6	62.2	68.2	69.8	67.8	58.4	44.1	24.0	0.4	(32)
(33)		10%	WB	-13.4	-0.9	21.2	35.1	44.5	53.6	57.3	55.0	45.8	33.4	16.7	-6.3	(33)
(34)		10%	MCDWB	-13.0	-0.1	24.8	44.4	57.9	66.5	68.1	65.3	54.9	40.0	19.0	-5.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.9	18.0	23.6	21.5	24.1	23.7	21.8	22.5	23.7	21.7	16.2	12.3	(35)
(36)		5% DB	MCDBR	16.9	20.4	25.1	28.4	32.9	31.4	29.1	29.8	32.0	27.3	20.4	16.8	(36)
(37)		5% DB	MCWBR	16.1	19.1	19.8	16.7	17.1	14.5	12.9	14.4	17.2	17.6	16.2	16.0	(37)
(38)		5% WB	MCDWR	16.7	20.4	24.8	26.4	30.8	26.2	24.3	25.6	28.1	26.6	19.2	16.7	(38)
(39)	5% WB	MCWBR	16.2	19.1	19.8	16.2	16.4	12.8	11.6	12.9	15.5	17.6	16.0	16.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.223	0.247	0.286	0.330	0.356	0.348	0.342	0.324	0.289	0.268	0.245	0.225	(40)	
(41)		taud	2.311	2.281	2.238	2.288	2.275	2.410	2.462	2.486	2.529	2.473	2.337	2.268	(41)	
(42)		Ebn at Noon	265	286	291	289	285	288	288	288	290	276	251	242	(42)	
(43)		Edh at Noon	19	27	35	37	40	35	33	30	26	22	19	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	443	768	1246	1715	1977	1914	1795	1555	1223	812	460	333	(44)	
(45)		RadStd	22	42	56	69	100	116	87	88	90	53	27	17	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page